Unit 7, Lesson 4: Interpreting Circle Graphs

Benchmark

MA.7.DP.1.4 Use proportional reasoning to construct, display, and interpret data in circle graphs.

Learning Target

☐ I can interpret the data in a circle graph.

7.4.1 Warm-Up: Would You Rather?

A 10-inch pizza with 8 slices and a 14-inch pizza with 10 slices were ordered for a party.

$10''$ **Pizza**	$14''$ **Pizza**
	

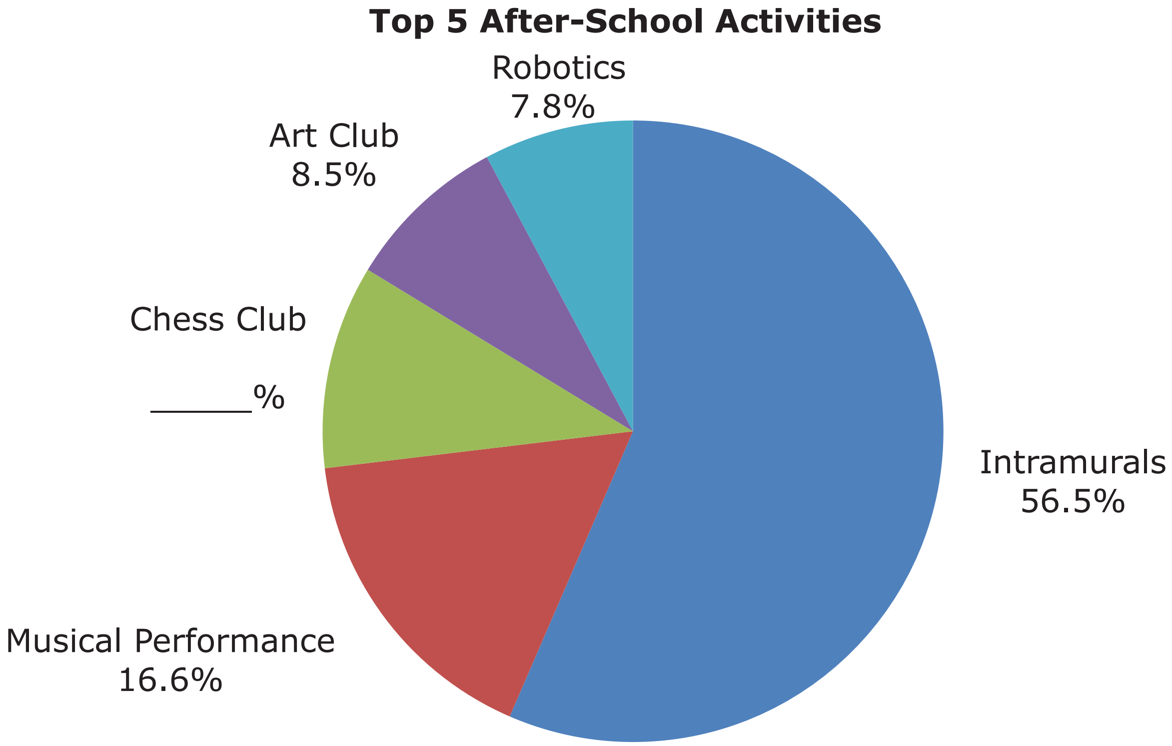
1. Would you rather have a slice from the $10''$ pizza, or a slice from the $14''$ pizza?

2. Explain your choice or justify it by showing your work.

7.4.2 Guided Instruction: Interpreting a Circle Graph

1. The graph reflects the top five after-school activities at the Capital City Recreation Center based on the number of participants for each activity.

   a. Fill in the missing percent for Chess Club.



   b. Which activity has the most participants?

   c. Which activity has the least participants?

   d. In which activity do about ½ of the kids participate?

e. In which two activities, when combined, do about $\frac{1}{4}$ of the kids participate?

f. Discuss with your partner how the circle graph helped you to answer the questions.

2. The Budget Committee members at the Capital City Recreation Center want to look at ways to better support their after-school programs. They decided to purchase one sketch book for every student in the art club.

a. Determine how many of the 283 total students that participate in extracurricular activities are in the art club by filling in the blanks.

Find 8.5% of 283.	8.5% of 283 ➡ 8.5% $\boxed{}$ 283 $8.5\% = \frac{8.5}{100} = \boxed{}$ $\boxed{} \cdot 283 = \boxed{}$ students in the art club

b. Determine how many of the 283 total students that participate in extracurricular activities are in the robotics club.

7.4.3 Collaboration: What Does the Graph Say?

Determine with your partner who will be partner A and who will be partner B. Partners A and B will be given two different ingredient lists for two popular holiday snack mixes and a blank circle. Do not let your partner see your list.

1. Use your recipe to fill in the chart. Then, construct a circle graph for your recipe using the circle provided. Round each percent to the nearest tenth and angle measure to the nearest whole number.

Your Recipe ______________________

Total Number of Cups _______

Ingredient	$\frac{\text{ingredient (cups)}}{\text{total cups}}$	Percentage	Angle Measure (°)

2. Exchange **only** your completed circle graph with your partner and use your partner's completed circle graph to answer the following questions.

   a. Which ingredient was used most in your partner's recipe?

   b. Which ingredient was used least?

c. Record your observations about any similarities or differences between your graph and your partner's graph.

d. Complete the table by using your partner's graph to calculate the number of cups necessary for each ingredient.

Partner's Recipe ___________________________

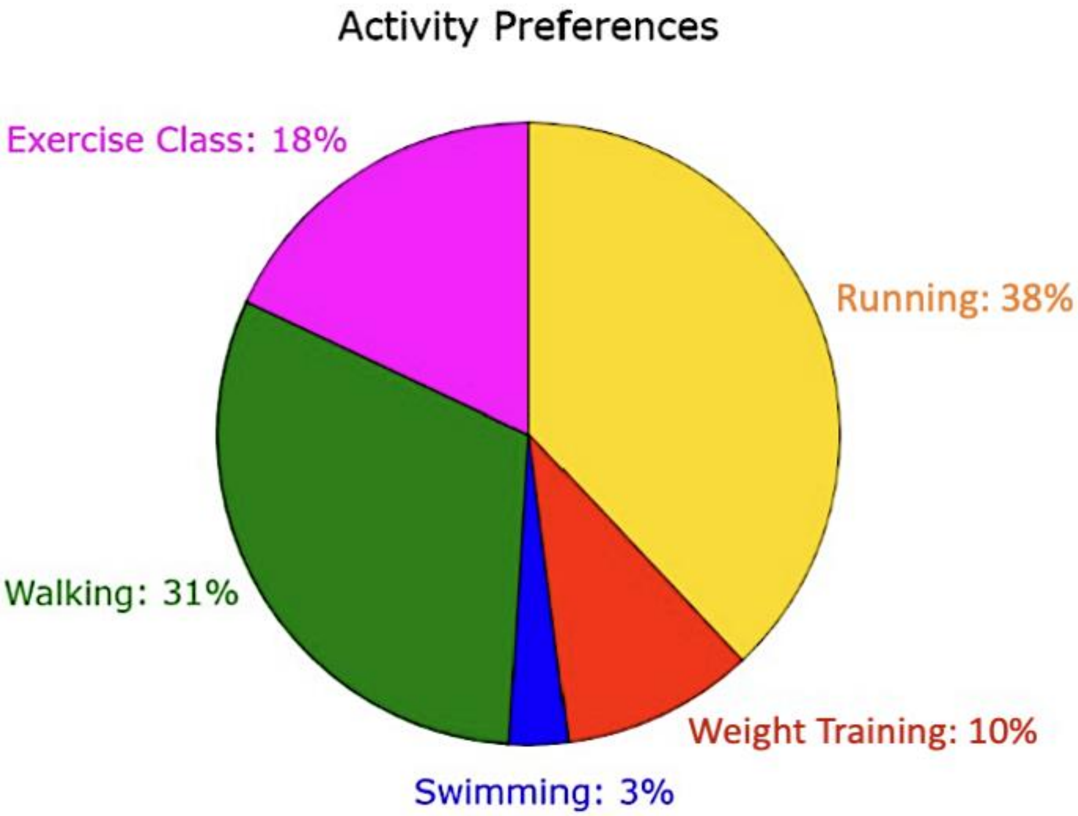
Total Number of Cups Used ________

Ingredient	Number of Cups

e. Explain the process you used to determine the number of cups for each ingredient in your partner's recipe.

7.4.4 Your Turn!

1. A health club surveyed 500 of its members to determine which activities they come to the club to participate in. Use the graph to answer the questions.



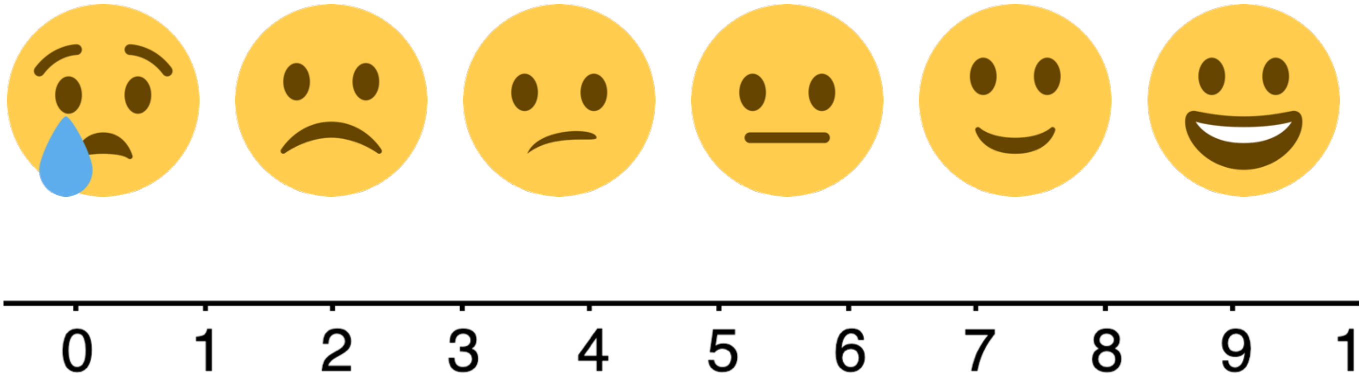
a. Describe how to determine which two activities approximately $\frac{2}{3}$ of the members participate in.

b. How many members come to the club for swimming?

c. How many club members participate in an exercise class?

7.4.5 Wrap-Up: Reflection

1. Choose the emoji that reflects your understanding of interpreting data in a circle graph.



2. Complete the statement.

   I chose this emoji because ____

   ____

   ____

   ____

